Did the Agent Reason Well?

Evaluating agentic forecasts — the answer and the reasoning

VECTOR INSTITUTE | INSTITUT VECTEUR

Ali Kore

Vector Institute | July 2026

What we'll cover

01

The prediction

A discrete rate call — cut, hold, or hike, four weeks out.

02

Score the answer

RPS, skill vs climatology, and an honest result.

03

Judge the reasoning

LLM-as-judge: right for the right reasons?

Predicting the BoC decision

A discrete event, not a time series: cut, hold, or hike at the next fixed announcement.

Forecast it 28 days ahead, as a probability over the three outcomes.

Outcome: ordered — cut < hold < hike

Lead: 28 days — T-1 just reads the yield curve

Why: four weeks out, the decision is still open

Read a single trace

Agent forecast · BoC · January 2025 origin

"The Bank is in a clear easing cycle, five consecutive cuts since June 2024. Inflation is near target and unemployment is rising..."

P(cut): 0.85 P(hold): 0.15 P(hike): 0.00

Cited: yield spread, unemployment, inflation

Trace: Langfuse f1945e8f... (the Bank did cut)

PART ONE

Score the answer

Proper scores for a discrete event

VECTOR INSTITUTE | INSTITUT VECTEUR

From CRPS to Brier to RPS

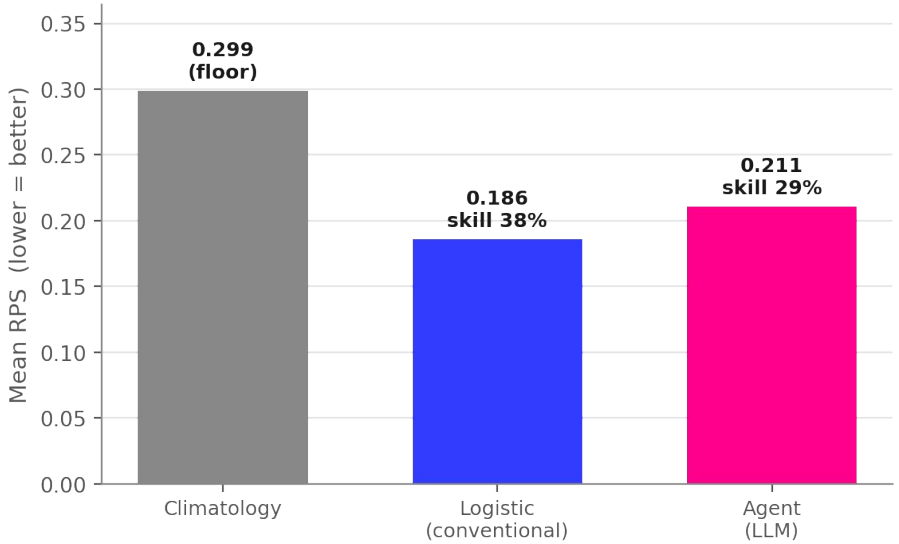
Yesterday's continuous forecasts used CRPS. A discrete event needs a discrete proper score.

Brier: binary — cut vs not

RPS: ordered — cut<hold<hike, distance-aware

Identity: RPS(K=2) = Brier; all strictly proper

Scored against climatology



Mean RPS · 12-meeting protected eval, 2025-26 · skill vs climatology

The conventional model wins.

Climatology (~76% holds) is the floor. Logistic beats it by 38%; the agent by 29%. The agent has real skill, and still loses to a logistic regression.

Why the agent trails

A default, un-tuned configuration, not a verdict on agents

Same four features

The agent sees exactly the logistic model's macro inputs, no search or code tools.

A single stateless call

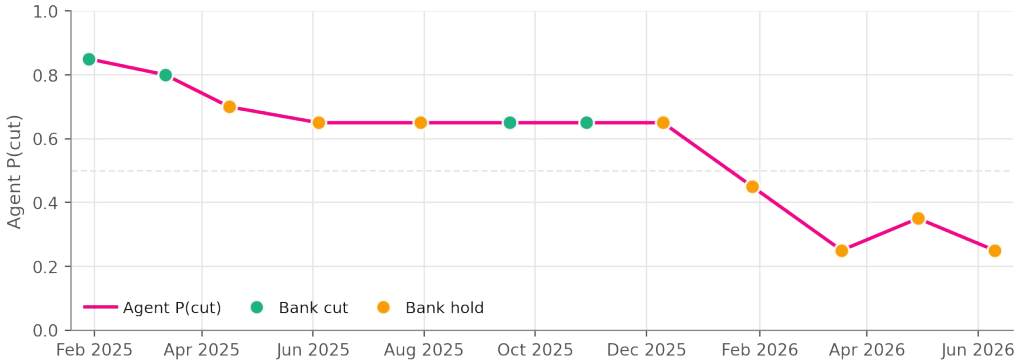
One analyst pass, fixed at config time. No curriculum, no learned strategy.

Absence of Dynamic Agent Tuning

The adaptive, self-improving agents were implemented primarily for the daily WTI task.

So the gap is a starting point, not a ceiling. The next two sessions close it.

What the score hides



Agent P(cut) vs the Bank's actual decision · 12 meetings

A cut-bias.

Through 2025 the agent kept 65-85% on a cut while the Bank held repeatedly (orange). Decent RPS, but a directional bias the number won't show you.

A good score is not a good forecast.

The agent can be right by luck, or wrong with excellent reasoning.

So evaluate the reasoning, not just the answer.

PART TWO

Right for the right reasons

Judging the reasoning, not the label

VECTOR INSTITUTE | INSTITUT VECTEUR

Evaluating an agent's process

Every run is a trace of steps (spans) we can attach scores to — two complementary kinds.

Quantitative: RPS, Brier — the label

Qualitative: LLM-as-judge — the reasoning

Substrate: Langfuse traces / spans / scores

An LLM-as-a-judge

```
verdict = judge_rationale_alignment(
    rationale=agent.rationale,
    key_signals=agent.key_signals,
    press_release_text=boc_release,
)
verdict.alignment_score    # 0..1
verdict.key_signal_overlap
verdict.justification
```

boc_rate_decisions/rationale_eval.py

Ground truth = the Bank's words.

Compare the agent's rationale to the Bank's published release for that meeting, cutoff-aware. The judge rates reasoning, not accuracy.

Right for the right reasons



12 meetings · correct vs aligned · real judge verdicts

Correct is not aligned.

Six right for the right reasons. But 2 were right for the wrong reasons, and 2 were wrong with reasoning that matched the Bank. The label misses four of twelve.

VECTOR INSTITUTE | INSTITUT VECTEUR

Two cases the label misses

WRONG CALL, ALIGNED

Jun 2025: said cut, Bank held

Alignment 0.85

Got jobs + inflation; missed tariff risk

RIGHT CALL, MISALIGNED

Mar 2026: said hold, Bank held

Alignment 0.40

Correct label; missed energy + geopolitics

The judge cites the Bank's own release.

Correct answers can hide bad reasoning.

Chain-of-thought is often unfaithful — the stated reason isn't the real cause.

> Evaluate the process, not just the outcome.

What to take forward

01

Score honestly

RPS, skill vs climatology, post-cutoff. The conventional model still won.

02

Judge the reasoning

LLM-as-judge vs the Bank's own release: right for the right reasons.

03

Process over outcome

Correct is not aligned. Evaluate the reasoning you'll deploy.

Open 03_rationale_alignment.ipynb to run the judge.